

Network Attack Detection & Reporting

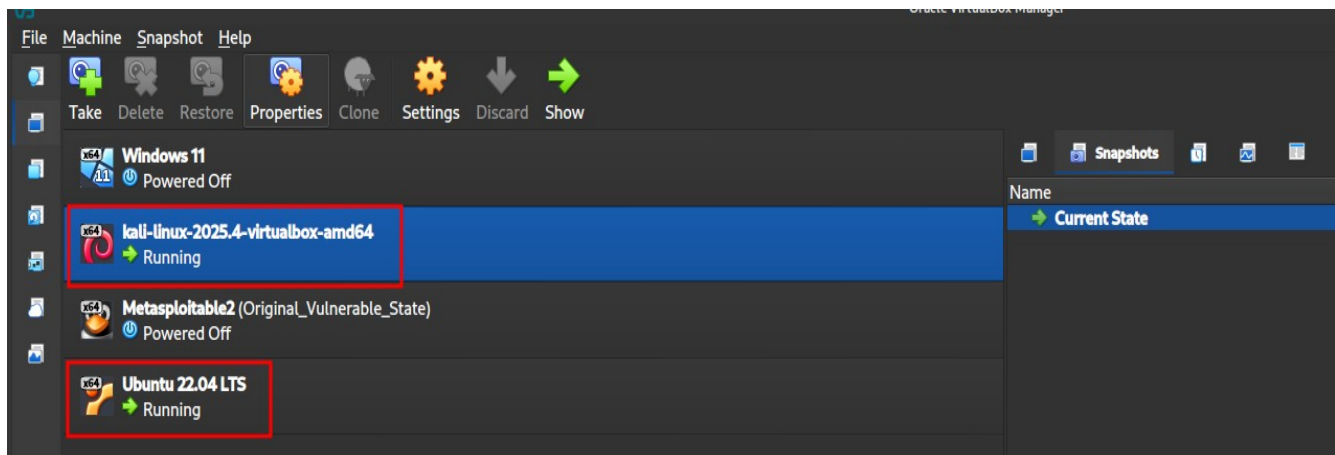
Introduction

The primary objective of this technical exercise was to simulate a real-world security breach within a controlled Security Operations Center (SOC) environment to evaluate threat detection and forensic logging capabilities. By deploying a Cowrie SSH honeypot on an Ubuntu Linux instance, the laboratory environment was configured to capture unauthorized interaction and network telemetry from a remote attacker using Kali Linux. This task involved the end-to-end execution of a cyberattack, the real-time capture of network traffic via Wireshark, and the extraction of granular session logs to demonstrate a comprehensive understanding of the incident response lifecycle. The following report details the findings, forensic evidence, and strategic mitigation efforts derived from this simulated intrusion.

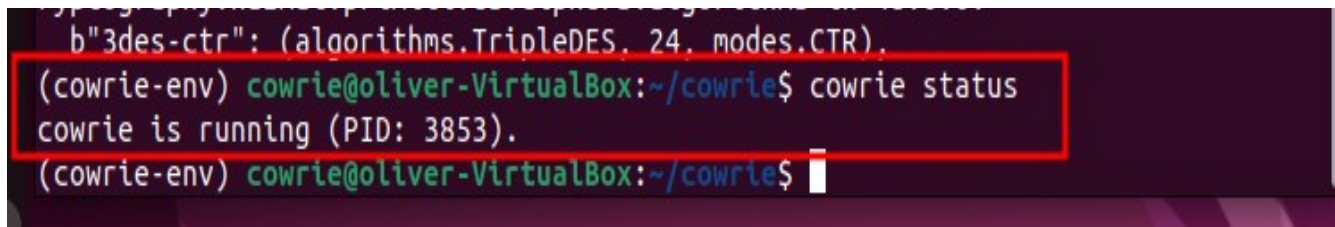
Hands-On

1. Environment Set-up

- i) Kali Linux VM - Attacker
- II) Ubuntu VM - Victim



2. Setting and Starting up Cowrie Honeypot



3. Identifying the Target (IP Address)

Run: ip a

```
(cowrie-env) cowrie@oliver-VirtualBox:~/cowrie$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b2:e8:3c brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
        valid_lft 317sec preferred_lft 317sec
    inet6 fe80::9f52:8396:4cf7:4b34/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(cowrie-env) cowrie@oliver-VirtualBox:~/cowrie$
```

4. Scanning the Honeypot (To see if it is active)

In Kali

Run: nmap -sV -p 2222 192.168.56.104

```
(kali@kali)-[~]
$ nmap -sV -p 2222 192.168.56.104
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-03 09:43 EDT
Nmap scan report for 192.168.56.104
Host is up (0.0013s latency).

PORT      STATE SERVICE VERSION
2222/tcp  open  ssh      OpenSSH 9.2p1 Debian 2+deb12u3 (protocol 2.0)
MAC Address: 08:00:27:B2:E8:3C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.93 seconds

(kali@kali)-[~]
```

5. Intrusion (SSH Login)

In Kali

Run: ssh root@192.168.56.104 -p 2222

```

(kali@kali)-[~]
$ ssh root@192.168.56.104 -p 2222
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
root@192.168.56.104's password:
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@svr04:~#

```

6. Execute the following commands

In Kali: whoami, ls -la, cat /etc/passwd, uname -a, Exit

Session Actions Edit View Help

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

root@svr04:~# whoami

root

root@svr04:~# ls -la

						Destination	Protocol
drwx	1	root	root	4096	2013-04-05 12:25	.	192.168.56.104 SSHv2
drwxr-xr-x	1	root	root	4096	2013-04-05 12:03	..	192.168.56.102 SSHv2
drwx	1	root	root	4096	2013-04-05 11:58	.aptitude	192.168.56.104 TCP
-rw-r--r--	1	root	root	570	2013-04-05 11:52	.bashrc	192.168.56.104 SSHv2
-rw-r--r--	1	root	root	140	2013-04-05 11:52	.profile	192.168.56.102 SSHv2
drwx	1	root	root	4096	2013-04-05 12:05	.ssh	192.168.56.104 TCP

root@svr04:~# cat /etc/passwd

```

root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh

```


9. Ubuntu log Analysis

a) Attacker's Fingerprint:

- i) Attacker's fingerprint : 192.168.56.102 (Kali VM)
- ii) Target Port: 2222
- iii) SSH Version: SSH-2.0-OpenSSH_10.2p1 Debian-3
- iv) HASSH Fingerprint: eeca2460550b9ded084ecf2f70a75356

b) Brute-Force Evidence

- i) **cowrie.login.failed**: An attempt at timestamp: 14:20:12 where the attacker tried the "username" root with the "password" root
- ii) This is a classic "low and slow" credential stuffing attempt

c) Session Duration:

The first session lasted **186.6 seconds**, while the second was **68.0 seconds**.